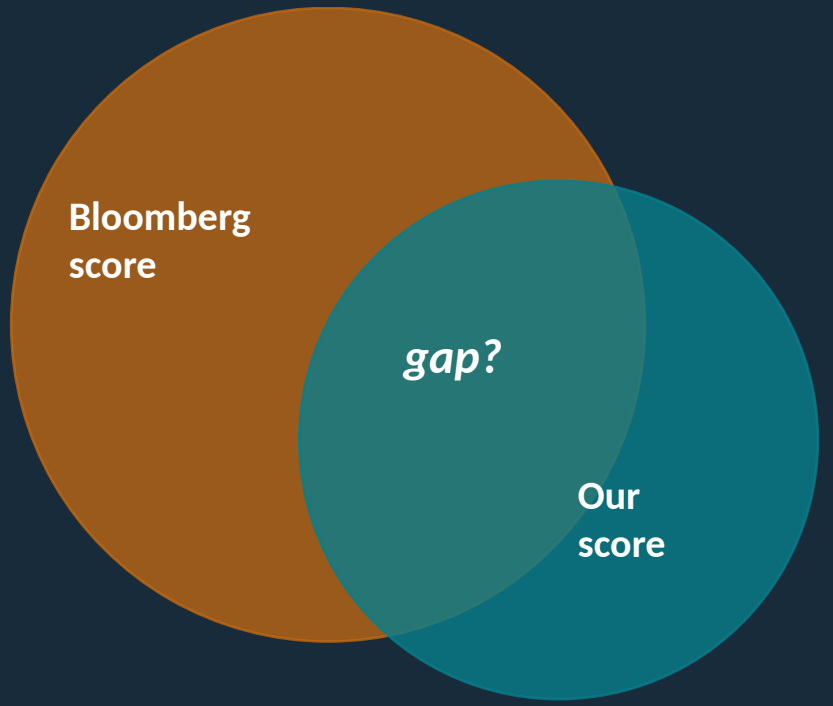
CHECKPOINT 1 | DATA AND EXPLORATORY ANALYSIS

Is the Vendor's Secret Sauce Priced?

Bloomberg vs a reproducible sentiment score on identical text



Neill Liran Botushanski

MMF 1941 · Proposal A16-02 · Supervisor: Professor Hamid · November 2026

Does Bloomberg's black-box sentiment add anything over a transparent score on the same headlines?

Q

Question

Does the gap between Bloomberg's daily news sentiment and an open, reproducible score on the same text predict the next return?

H

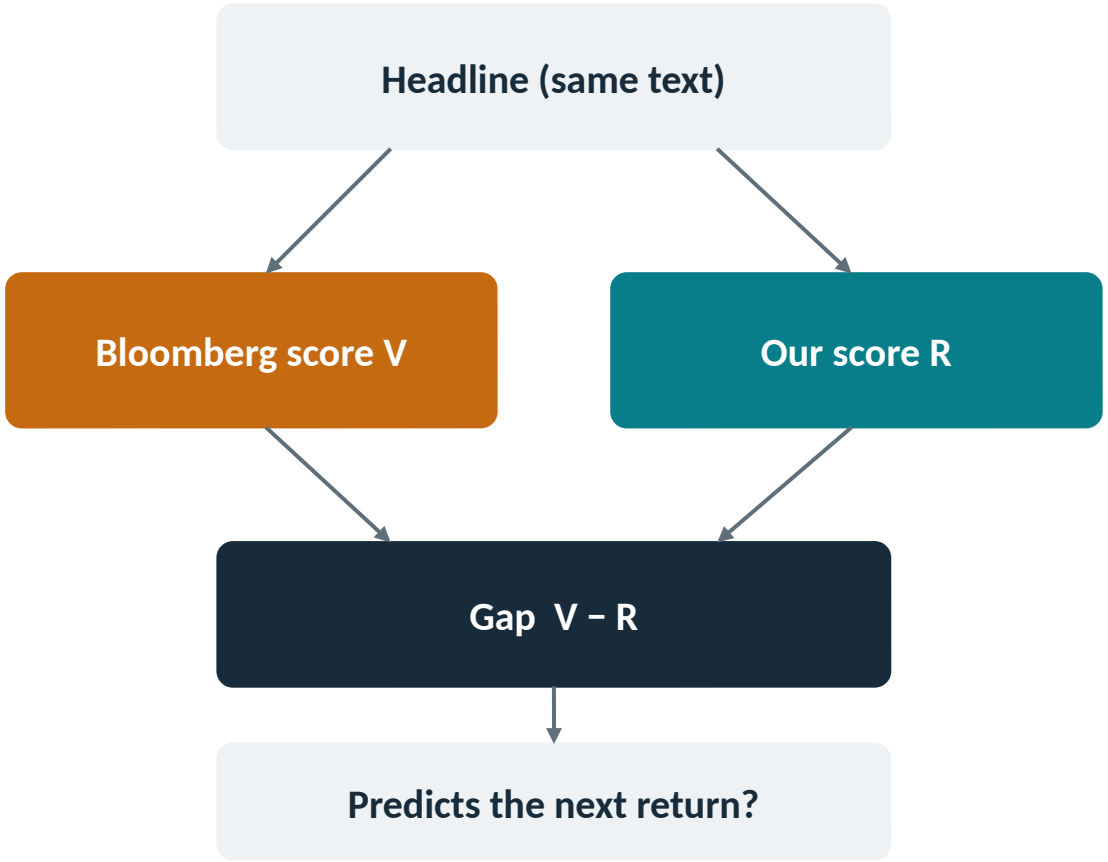
Hypothesis

The gap is mostly noise: it adds nothing robust beyond the reproducible score (an honest null on a "black-box premium").

!

Why it matters

If a free, auditable score does as well, the vendor's secret sauce is not priced, and a costly input can be replaced.



One pre-specified test: does the gap predict returns beyond the reproducible score?

Primary specification

$$r(i, t+h) = \alpha + \beta \cdot R(i,t) + \gamma \cdot [V(i,t) - R(i,t)] + \text{controls} + \varepsilon$$

H0: $\gamma = 0$ (no black-box premium)

Estimation: HAC / clustered standard errors; encompassing and Diebold-Mariano tests

Robustness: purged cross-validation, reproducible-only vs + vendor features, deflated Sharpe ratio

Design element	Setting
Unit	Security-day
Universe	Point-in-time S&P 500 members
Sample	2020-2025 daily (buffers Jul 2019, Jan 2026)
Outcome	Next-day or short-horizon return
Text	188 headlines (identical text, D1)
Secondary	Vendor score on the full panel (separate exercise)
Still open	Horizon and scoring rule (E2)

Seven data components: Bloomberg supplies six, RiskLab the text

#	Component	Role	Source	Status
1	**Vendor sentiment score**	The black box being tested	Bloomberg NEWS_SENTIMENT_DAILY_AVG	**In hand**
2	**Story counts**	Coverage checks, controls	Bloomberg NW043-NW046	**In hand**
3	**Headline text**	Input to our score	RiskLab News_Articles.csv (188)	**In hand**
4	**Returns**	Outcome	Bloomberg total-return index, open, close	**In hand**
5	**Point-in-time universe**	Who is in the sample each day	Bloomberg INDX_MWEIGHT_HIST, 73 month-ends	**In hand**
6	**Identifiers and controls**	Joins, size, liquidity, sector	Bloomberg BDP, market cap, volume, BICS	**In hand**
7	**Earnings dates and index**	Event control, market return	Bloomberg earnings history; SPX, SPXT, VIX	**In hand**

Cross-checks only: 3 RiskLab vendor-sentiment files, still to copy from the RiskLab drive. **Intraday data:** none requested from Bloomberg (brief rule).

RiskLab supplies the headline text: 188 headlines on 19 firms, none from Bloomberg

188
headlines, 2020-2025

19
large-cap firms

95
public source domains

0
from bloomberg.com

- Dates only, no publication times
- Hand-collected, mostly ESG themes (Reuters 32, CNBC 11, Guardian 7, ESG Today 7)
- 3 URL-date mismatches and 1 missing URL: flagged, kept, reported with and without
- RiskLab holds no larger Bloomberg headline archive (Professor Hamid)

Decision D1

No Bloomberg headline export. These 188 headlines are the primary identical-text sample: the supervisor's fallback, with a conservative return cutoff. To confirm with him at this checkpoint.

The terminal pull covers 635 tickers over six and a half years and passed every pilot check

635
exchange tickers = 617 companies

73
month-end membership snapshots

978,472
security-days of prices and returns

1,070,056
security-days of news fields

7 / 7
pilot checks match the terminal (HP, GP)

26
fields confirmed in FLDS

Pulled 2026-11-06 on the RiskLab terminal, end-of-day only. The sentiment value dated D covers the 24 hours to 09:20 New York on D (FLDS).

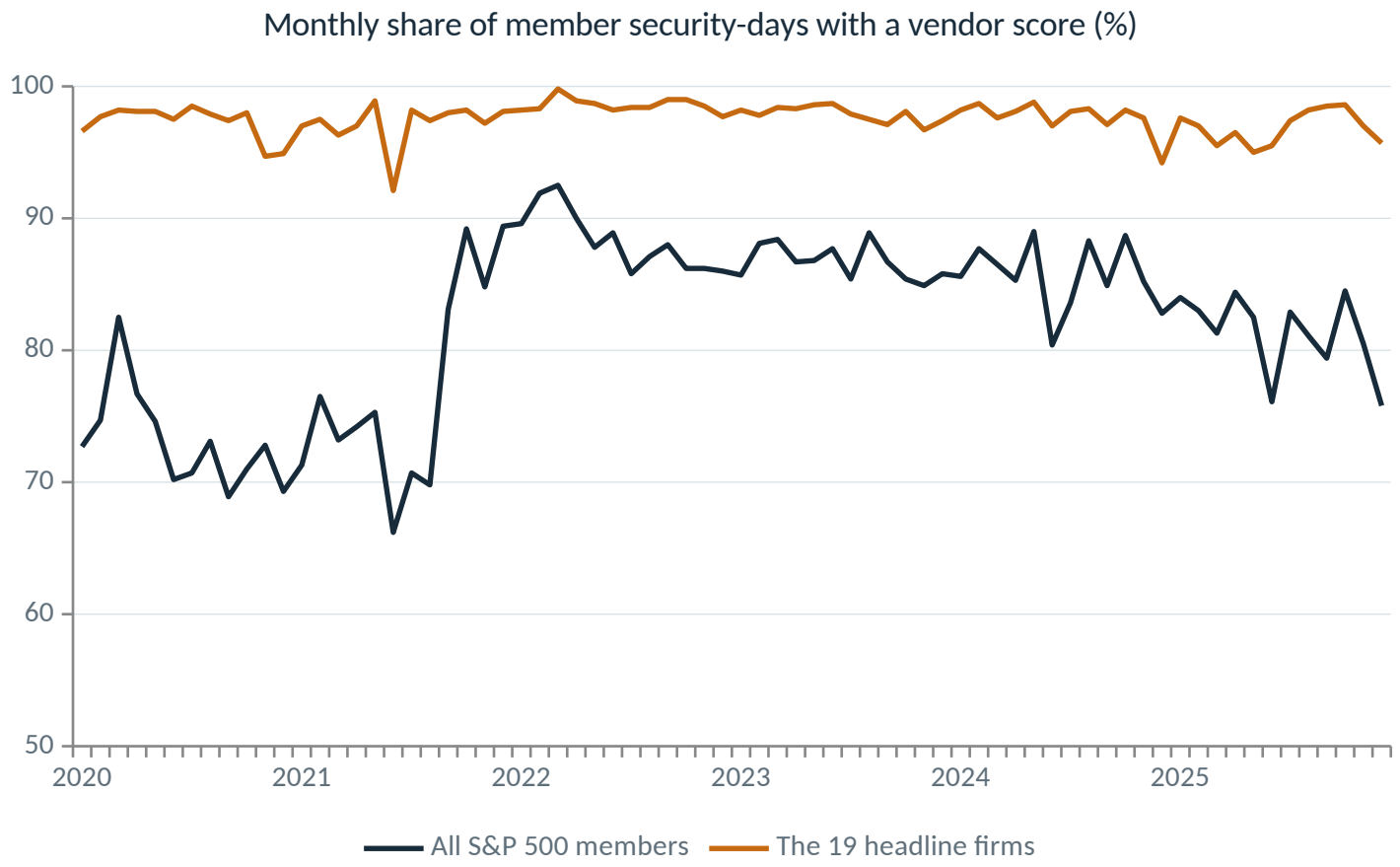
Every step from the terminal to the panel leaves an audit trail

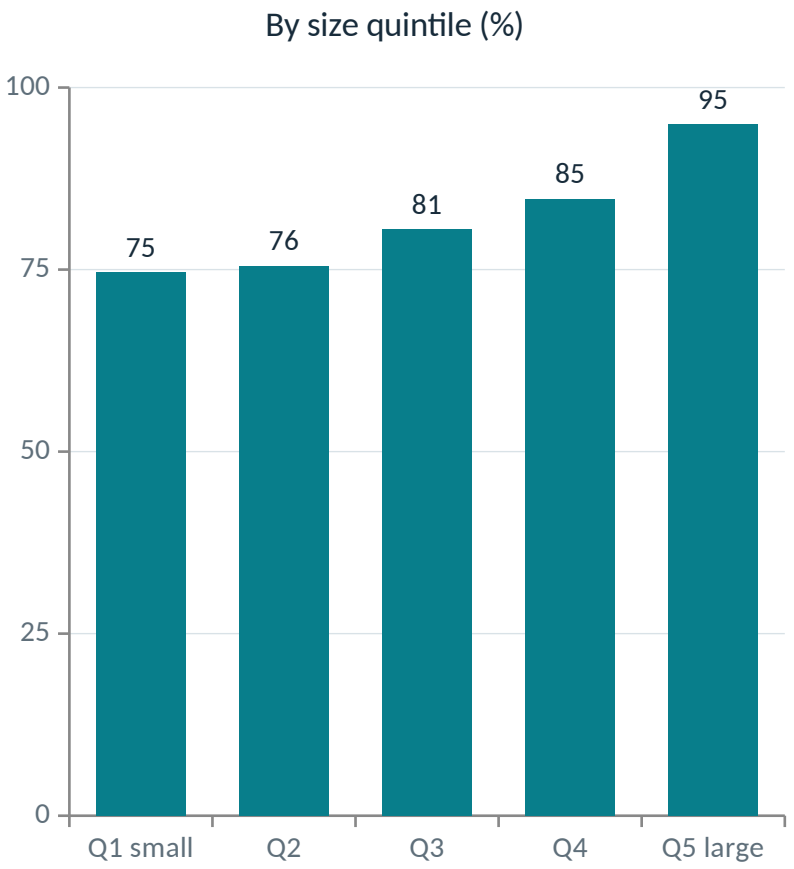
1. **Formula workbook**
   BDS, BDH and BDP formulas, built by a script
2. **Field check + pilot**
   26 fields, 7/7 checks
3. **Terminal pull**
   2026-11-06, no data-limit errors
4. **Copy and hash**
   11/11 hashes match
5. **Export + source log**
   tidy CSVs, 42 log rows
6. **Validate**
   76 checks, 4 flags explained

Rules kept throughout

Raw files are never edited; every derived file lives in data/interim or data/analytic. Each script reads declared inputs, writes declared outputs and a summary, and the source log carries every file's SHA-256.

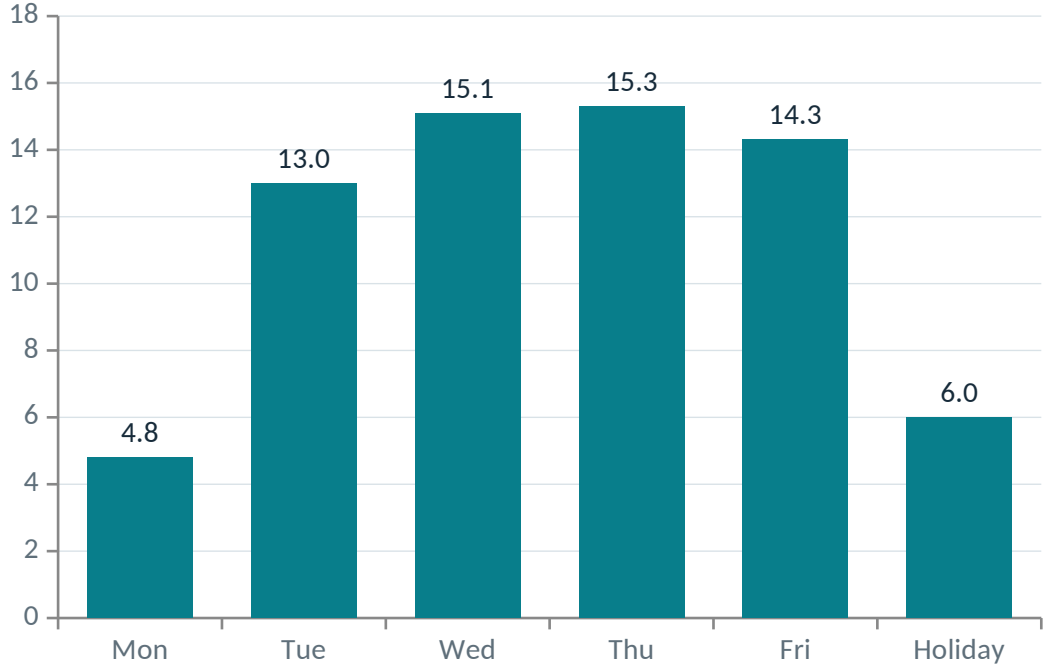
82% of member security-days carry a vendor score, and coverage rises with firm size



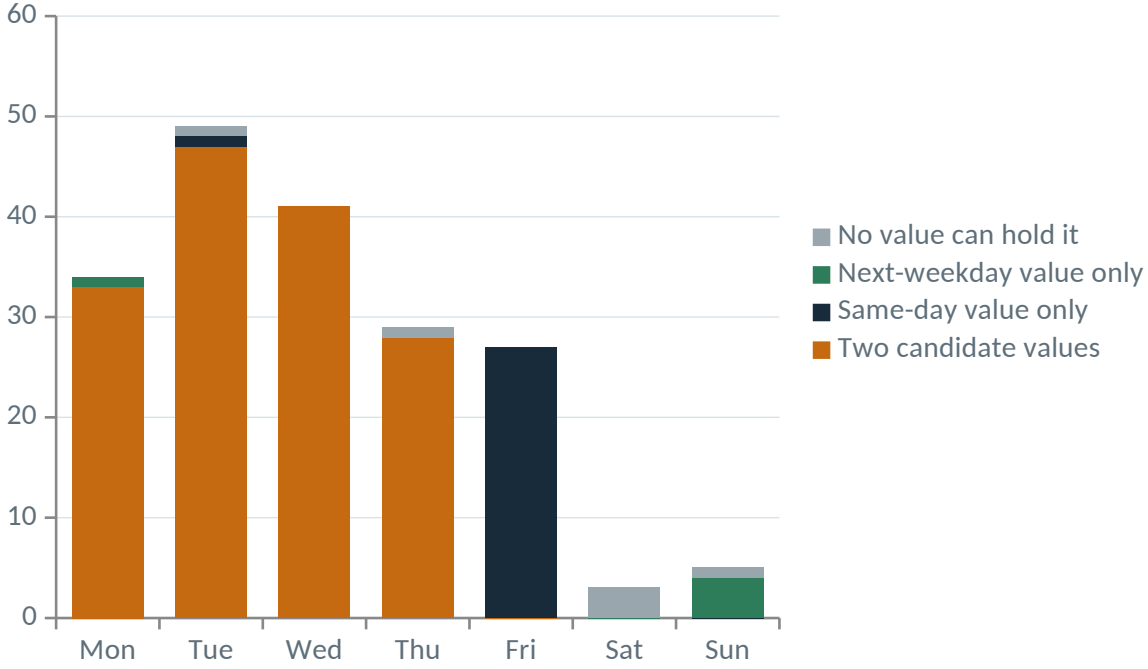


Monday and holiday values hold a third of a normal day's stories: a weekend gap in the vendor data

Scored stories per security-day, by value date



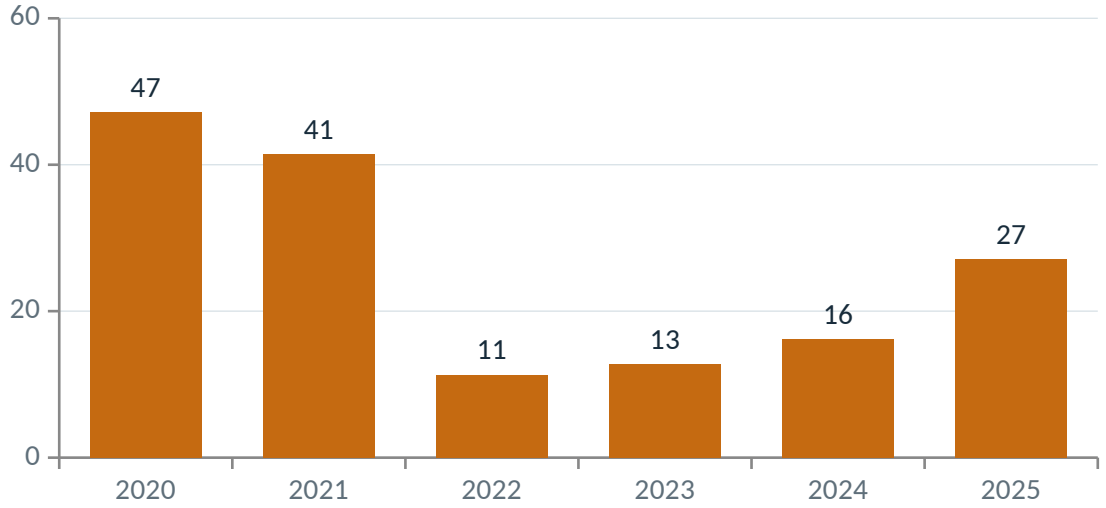
Headlines by weekday: which daily value could hold the story



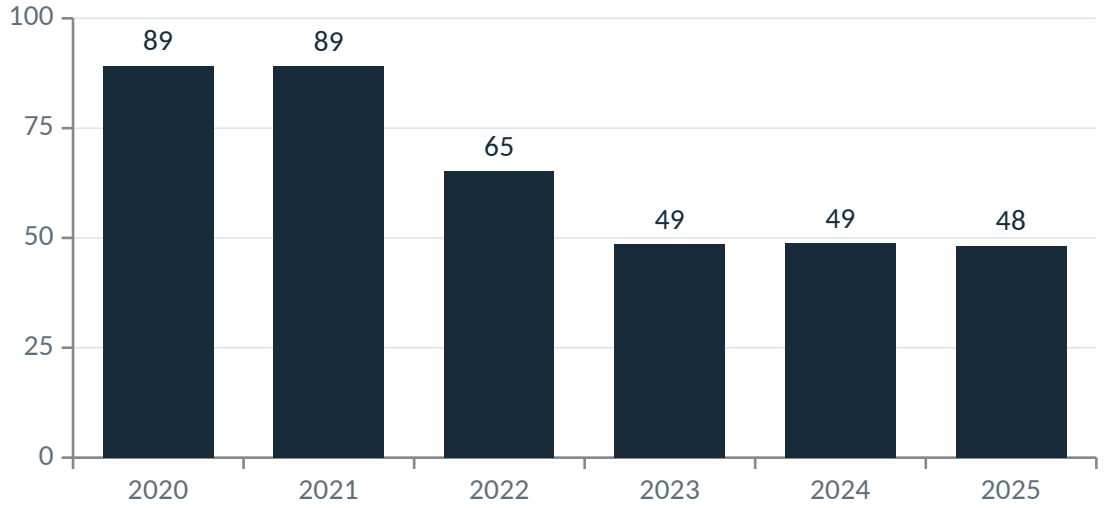
A value dated D holds the stories of the 24 hours to 09:20 New York on D, and none is dated Saturday or Sunday. Headlines carry dates only, so 149 of the 182 with a vendor value could sit in two different daily values.

The vendor score changes regime around 2022, so it must be rescaled inside each training window

Share of vendor scores exactly 0 (%)



Mean stories per security-day



0.81
correlation with (positive − negative) / scored stories

25%
of all values exactly 0: real neutral readings, not missing

0.48
standard deviation in 2022, the peak year

Every observation lost between the universe and the analysis samples has a recorded reason

Step	Rule	Count
1	Universe: members at the latest month-end before each trading day	759,846
2	… with a Bloomberg market row	759,347
3	… with a close price	759,228
4	… with both same-day returns	759,225
5	Universe days with a vendor news row	759,303
6	… with a vendor sentiment value	623,105
7	… and a same-day open-to-close return	622,618
8	… and a next-day return (broad sample)	622,570
A1	Headlines dated 2020-2025	188
A2	… firm in the index that day	186
A3	… a vendor value on a candidate date	182
A4	… vendor values on both dates (ambiguous)	149
A5	… a return after the conservative cutoff	186
A6	… without a flagged date or URL	182

76 checks

4 flags, all explained

0

duplicate keys or many-to-many joins

499

universe days with no price: 497 delisted mid-month, 2 not yet listed

One command rebuilds every output, byte for byte, from the raw files

Module	Step	Writes
02	**Raw validation**	76 checks; file hashes
03	**Signals**	vendor values, headline alignment
04	**Outcomes**	open-to-close and total returns
05	**Panel and attrition**	point-in-time panel, join report
06	**Descriptive diagnostics**	coverage tables, figures
07	**Hand checks**	raw cell to panel variable
—	**Report and this deck**	notebook, HTML, PowerPoint

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hand-checked observations, raw workbook cell to panel

7 / 7

terminal checks (pilot)

```
python src/run_all.py
```

Six questions for Professor Hamid before estimation

D1 No Bloomberg headline export: are the 188 headlines acceptable as the primary sample?

P5 Which vendor value should each headline be compared with: next weekday, same day, or their average?

E2 Primary horizon: open-to-close on the value date, or next-day close-to-close?

E2 Primary scoring rule: Loughran-McDonald lexicon or FinBERT, with the other as robustness?

O6 Keep Friday and Saturday headlines, given the weekend gap?

S3 Start the secondary exercise on the broad panel now, or after the design freeze?

Next checkpoint

- Freeze the design log (P1-P6, E1-E2)
- Build both reproducible scores on the 186 member headlines
- Run the primary test: HAC / clustered SE, encompassing and DM
- Secondary: vendor score on 622,570 security-days

Appendix: every data source and its validation status

Dataset	What it holds	Status
BBG-P2-UNIVERSE	S&P 500 members at each month-end (INDX_MWEIGHT_HIST)	AVAILABLE; validated (02)
BBG-P2-SECURITIES	Identifiers, status, currency, GICS (current), composite FIGI, BICS	AVAILABLE; validated (02)
BBG-P2-MARKET	Open, close, total-return index, volume (exchange and composite), market cap	AVAILABLE; validated (02); 10 hand checks (07)
BBG-P2-NEWS	NEWS_SENTIMENT_DAILY_AVG, story counts (all, pos, neg, neutral), NEWS_HEAT	AVAILABLE; validated (02); 10 hand checks (07)
BBG-P2-EARNINGS	Earnings announcement dates, times, EPS	AVAILABLE; validated (02)
BBG-P2-INDEX	SPX open/close, SPXT, VIX	AVAILABLE; validated (02)
BBG-P2-RAW-WORKBOOK	Raw pull workbook (formulas + cached values)	AVAILABLE; hash matches terminal record
RL-114CB44BEECC	Headlines: Company; Date; Headline; URL	AVAILABLE; copied byte-for-byte 2026-11-08
RL-1EBA802A51B7	Vendor sentiment (NW020) and call implied vol (RK074) for part of the sample	PENDING COPY (2.9 MB)
RL-4CCDB3AF35B1	Vendor sentiment, wide	PENDING COPY (3.8 MB)
RL-6E2DA04334F1	Includes NEWS_SENTIMENT_DAILY_AVG and story counts per S&P name	PENDING COPY (95 MB)
FRD-STOCKS-1MIN	One-minute stock bars	AVAILABLE per inventory; not copied